Assignment: AMIS1003 Introduction to Cybersecurity

CLO3: Discuss on communication security issues that can arise from using e-mail, social networks, voice over internet protocol, instant messaging and mobile devices. (C3, PLO6)

Title: Introduction to Ransomware

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TITLE: RANSOMWARE

Introduction

Definition

Ransomware is a type of malware software which prevents you from accessing your device and the data stored on it, usually by encrypting your files. This form of malware has expanded into a top cybersecurity threat globally.

Evolution of Ransomware

The first ransomware was discovered in the late 1980s and was called the "AIDS Trojan" or "PC Cyborg." It spread by floppy disks and locked access to files until payment was made.

In 2013, a new ransomware called CryptoLocker appeared. It used strong encryption to lock files and required for payment in Bitcoin or online vouchers. The group behind CryptoLocker made about US $3 million before police stopped it in 2014 during Operation Tovar.

In May 2017, another ransomware called WannaCry spread very quickly using a Windows security flaw known as EternalBlue. It is affecting over 230,000 computers in 150 countries and targeting entities such as Telefónica in Spain, and FedEx in the United State Each victim was asked to pay about US $300.

At the same year, a Petya ransomare variant hit Ukraine and other countries. It did not only encrypt files but also damaged the system's startup process, which made computers unusable. The attack seems designed to cause disruption rather than earn money. Ransomware developed quickly into new forms.

One big change was Ransomware-as-a-Service (RaaS). In this system, expert hackers can build ransomware and rent it to other criminals, which is affiliates who use it to attack and then share the money with the creators. This would made the ransomware easier to spread and more common. For example, Conti ransomware which is first seen in 2019. This ransomware was created by the Russian group called "Wizard Spider." Conti spread through Windows networks, locked files, and gave attackers control after infection. Because of the damage it caused, the U.S. government offered a US $10 million reward for information about the operators. Recently, ransomware has become common and damaging. In 2024, around 59% of organizations were hit. It is because the average ransom payment increased from about US $400,000 in 2023 to US $2.73 million in 2024. Although, ransomware payments reached US $813 million that year.

In early 2025, only 17% of companies in the UK paid for ransoms. It is the lowest ever because many of them had better backup systems to recover their data without paying money. Right now, important services and infrastructure are still main targets for ransom. In 2024, ransomware made up nearly half of all cybercrime reports to the FBI, with total losses rising to US $16.6 billion. In early 2025, these attacks increased by 49%, especially against US companies in manufacturing, construction, and IT. This growth was linked to Ransomware-as-a-Service (RaaS) and weaknesses in remote work systems. Many organizations use backups to protect against ransomware. It is because sometimes these backups can store offline or made unchangeable, so that the attackers cannot damage them. Because of this, some victims can

recover their files without paying. Paying ransom does not guarantee success. Only about 46% of victims got their data back, and much of it was broken. Recovery is also slow and expensive.

In average, ransomware causes about three weeks of downtime, and the costs can reach millions. Services like hospitals and government offices are among the most at risk. Nowadays, attackers use tougher methods. One of the common methods is double extortion. They not only lock files but also threaten to release stolen data. Another method is pure extortion, where they skip the file encryption and only use the stolen data as a threat. Overall, ransomware involve crime, fast spread, and advanced technology. It started as simple attacks but has now developed into modern forms like Ransomware-as-a-Service and double extortion. However, it can cause high costs, risks to important systems, and major disruptions. As we know ransomware is one of the biggest cybersecurity problems today and we must know its history, methods, and impact.

Online Communication Security Issues of Ransomware

Ransomware is one of the most dangerous online security problems today. It is a type of malicious software that blocks access to important files or entire systems and then demands money to restore them. One of the main ways ransomware spreads is through the weakness in online communication. Attackers usually use email, messaging apps, remote access tools, and even phone calls to trick users to give them access.

Email is the most common method for spreading ransomware. Attackers send phishing emails that look like they come from trusted sources, such as banks, delivery companies, or work colleagues. These emails usually include an attachment or a link. When the user opens the file or clicks the link, the ransomware gets installed on the system.

Email systems are also weak because many organizations do not use basic security standards. These are the tools that check if an email really comes from the domain it claims to be such as DomainKeys Identified Mail (DKIM). Without them, attackers can send fake emails that appear real, which makes it easier for people to fall into the trap.

Chat apps like Microsoft Teams and WhatsApp are also used to spread ransomware. These apps can let peoples share files quickly. Attackers also can take advantage to send dangerous files. If someone downloads the file, the ransomware can infect their computer and spread to others.

Even though these apps are used encryption to protect messages, but it does not help if the user clicks on something harmful. The attack happens on the device itself. Apps may be safe, but they can become risky when users are not careful with what they open.]

Another example is through phishing emails, where the attackers use emails that look like legitimate emails sent by their friends or colleagues. But in truth, they're infected by ransomware that infiltrates the system once the user clicks on the email.

Besides that, poor password creating and lack of multi-factor authentication can compromise online communication channels. Attackers can use this chance to distribute ransomware within a week large organization's network. Many businesses also rely on cloud-based platforms for exchanging documents.

Without proper security configurations, the platforms can be vulnerable to ransomware, especially when files are shared to multiple users.

Why Does Ransomware Happened

Ransomware is one of the most common cyber threats today. It is a type of malicious software that blocks access to files or systems until a payment is made. This issue affects individuals, companies, and even governments.

The main reason ransomware attacks happen is financial motivation. Criminals use ransomware to force victims into paying money, usually in digital currencies like Bitcoin. Digital currencies make it hard for the police to trace the attackers. According to Richardson and North, many ransomware groups will make millions of dollars every year from these payments. The effect is victims will feel pressured to pay the money to recovery their data.

Another reason is the low risk for attackers. Cybercriminals can work from anywhere in the world. They do not need to be near their victims, which makes it harder for police to catch them. This encourages more criminals to try ransomware because the rewards are high while the risks are low.

Besides that, ransomware happens is because attackers use tricks like phishing and social engineering. Phishing is when criminals send fake emails or messages that look real to fool people into clicking links or opening files.

Social engineering is when attackers trick people into giving away information or access without knowing it. Both methods are common ways to fool people, which makes it easier for criminals to spread ransomware.

Another important reason ransomware is successful is poor backup practices. Many individuals and organizations do not create regular copies of their important files. When ransomware locks their data, victims feel forced to pay because they have no other way to recover their information. If proper backup systems existed, the damage from ransomware would be less severe.

However, without backups, attackers gain more power and victims become more desperate to pay the ransom.

How does Ransomware Happened

One of the most common ways to let ransomware occur is through phishing emails. These emails are designed to look very similar. It is often pretending to be from banks, insurance companies, and other service providers. They include attachments that are disguised as invoices, reports, or links that lead to fake websites. When the victim clicks or opens the file, ransomware will silently install. Since phishing depends on social engineering rather than technical flaws, it remains one of the easiest and most effective attack methods.

Another frequent method is through malicious websites or drive-by downloads. Attackers create the websites and online advertisements so that ransomware can be installed automatically. If the victim's system has not been updated with important security patches, the malicious page can trigger the ransomware download. This is especially dangerous for users who delay updates. It is because old systems often have weaknesses that attackers already know how to use.

Ransomware can also spread through infected software, especially cracked programs. Attackers hide ransomware in the illegal downloads because they know people who download

free software usually do not check whether if it is safe. When victims install programs, the ransomware gets into their systems without them realizing it.

Another way to let ransomware happen is Remote Desktop Protocol (RDP) attacks. RDP is a tool that lets people control a computer from another location. If the password is weak or stolen, attackers can log in. Once attackers log in, they can install ransomware, switch off security tools, and spread the infection to other computers on the network.

Besides that, attackers also exploit weaknesses in operating systems, apps, or network devices that have not been updated. They use scanning tools to search for these unprotected computers. Once they find a weak system, they will quickly install ransomware. This method can let them infect many devices without the user doing anything.

Once ransomware gets into a network, it can spread like a worm. It moves from one computer to another, often by using stolen passwords or unpatched software. In companies with shared drives, this can cause huge data loss in just a few hours.

Finally, ransomware can spread through removable media like USB drives or external hard drives. When these infected devices are plugged into a computer, the ransomware will install itself. This method is less common today, but it is still used in some targeted attacks.

Countermeasure And Prevention of Ransomware

The growth of the internet has brought many benefits, but it has also created a range of hidden threats. Ransomware is one of the most dangerous and disruptive types of cybercrime. This harmful malware encrypts or locks important files and systems, then demands payment, often in cryptocurrency, to restore access.

These attacks can lead to significant financial loss, disrupt business operations, and even threaten critical infrastructure. Firewalls, antivirus programs, intrusion detection systems, and endpoint protection offer multiple layers of defence, stopping many threats before they spread. When it combined, these tools can detect unusual activity, which is block the malicious files from executing and limit the impact of the attack. For example, application whitelisting will ensure that approved software can run, email filters will lower the chance of phishing emails reaching users' inboxes.

These layers make it harder for ransomware to succeed. Human awareness is another important defence. Many ransomware infections start with phishing emails, deceptive links, or unsafe downloads.

Training users to recognize suspicious messages, attachments, and social engineering tricks can greatly reduce the chances of an attack. In addition to training, organizational strategies like segmenting networks, limiting access rights, and enforcing multi-factor authentication can reduce damage if attackers breach one part of the system.

At the same time, the best way to defence against ransomware is a proactive, layered approach. By combining updated systems, strong security tools, secure backups, and well-informed users, we can build resilience against even the most serious threats.

By promoting a culture of cybersecurity awareness and readiness, individuals and organizations can significantly lower their risk and make ransomware attacks less likely to succeed. Network segmentation is another method to protect against ransomware. Instead of linking all systems in one large network, segmentation splits them into smaller and isolated sections.

If ransomware enter one area, it is less likely to spread to others. There are some critical systems such as databases or backups can place in more secure segments with stricter access controls. This can limit the attacker's movement and reduces potential damage. It also giving administrators more time to respond and contain the threat.

Keep systems up to date is another critical step. Attackers often exploit old vulnerabilities in operating systems or applications that have already been fixed by software vendors. If updates are ignored, the same weakness can be used repeatedly. Automating patches reduces the chance of human error and closes security gaps faster.

Another important countermeasure is using strong password policies and multi-factor authentication. Many ransomware groups break into systems by guessing weak passwords or stealing login details. With strong, regularly updated passwords and multi-factor authentication, attackers would spend a lot of time to break in, even if they already know a password.

According to Verizon, stolen or weak credentials are one of the main ways cybercriminals break into systems, so stronger authentication greatly reduces risk. Ransomware is a worldwide issue. Attackers often work across different countries, which makes it hard for police to stop them. To counter ransomware, governments, companies, and cybersecurity organizations can share some information about ransomware and work together.

At the same time, international cooperation can disrupt some criminal groups and take down their servers. Governments should held awareness campaigns can educate the public on how to stay safe and reduce the risk of attacks. They can upload some infographic in media social such as Facebook and TikTok.

One more way to counter ransomware is using threat intelligence and monitoring services. These tools can collect real-time data about new ransomware variants, attack methods, and criminal activities across the world. By analysing this information, security teams can update defences quickly, prevent harmful actions, and prepare for emerging threats before they strike.

For example, intrusion detection systems can use updated threat intelligence to spot unusual patterns at an early stage. This proactive approach focuses on stopping attacks before they happen, which lowers the chance of success and reduces the damage of ransomware.

Conclusion

In conclusion, ransomware is still one of the biggest problems in the online world. First, ransomware was just a simple virus that asked for money. Today, it has grown into a more advanced crime.

Criminals now use strong encryption to lock files, making it harder to stop and more harmful for victims. The damage from ransomware is very serious. The cost is victims might

lose money not only by paying ransom but also by fixing systems, hiring experts, and losing time when their work stops.

Besides that, companies may lose the trust of their customers and face legal issues under data protection laws. Fighting ransomware requires both technology and human action. From the technical side, systems should always be updated, and strong security software must be used.

From our side, employees need training to recognize suspicious emails, links, and files so that they do not become an easy target for attackers.

Besides that, workers must learn how to spot phishing emails and strange activities. Many attacks succeed because people click unsafe links or download harmful files. The question of whether to pay ransom is still a big problem. Some victims decide to pay the ransom because they feel hopeless, but this only motivates criminals to keep attacking. Also, there is no guarantee that paying can return the data safely.

Now, ransomware also becoming more advanced. For example, hackers use double extortion to lock the files but also threaten to release private information. Some groups even sell a ransomware tools called Ransomware-as-a-Service which makes it easier for more criminals to join these attacks.

Governments, companies, and security groups should work closely together since ransomware has become a global problem. They can share threat intelligence and exchange information about new attack methods and ways to block them. This cooperation is very important because criminals often operate across countries, so no one can stop them alone.

In short, ransomware shows the fight between criminals and defenders. It mixes technology, human mistakes, and weak systems. The solution is not one single action. It needs constant care, which is by strong security, trained people, good planning, and teamwork across countries.

Only with this kind of teamwork can we make ransomware less dangerous and help systems recover more quickly after an attack.

(2741 words)

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